PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
from sklearn.neural_network import MLPClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
X = [[25, 110, 85, 170], [60, 150, 160, 250], [35, 120, 95, 190], [55, 145, 150, 240],
    [28, 115, 90, 175], [65, 160, 170, 260], [40, 125, 100, 200], [58, 155, 165, 255],
    [30, 118, 88, 180], [62, 152, 158, 245], [45, 130, 110, 210], [52, 148, 155, 238],
    [22, 108, 82, 165], [68, 165, 175, 270], [38, 122, 98, 195], [57, 150, 162, 248],]
y = [0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=2, stratify=y)
scaler = StandardScaler().fit(X_train)
X_train, X_test = scaler.transform(X_train), scaler.transform(X_test)
model = MLPClassifier(hidden_layer_sizes=(8, 4), activation="relu",
                      max_iter=2000, random_state=2)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Test accuracy:", accuracy_score(y_test, pred))
patient = scaler.transform([[50, 140, 140, 230]])
result = model.predict(patient)[0]
print("New patient ->", "Disease likely" if result == 1 else "No disease")
```

OUTPUT:

```
>>>
========== RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ==========
Test accuracy: 1.0
New patient -> Disease likely
>>>
```

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